

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7

Name: Type your name

Date: Today's date

Class: Class period



TODAY'S OBJECTIVES

Content Objective: I can use ratio reasoning to convert measurements between the customary and metric systems.

Language Objective: I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion factor

Customary system

Metric system

Approximately

Ratio table



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

The ratio that trades one unit for another is a

2

Inches, pounds and gallons belong to the

system.

3

Centimeters, grams and liters belong to the

system, built on tens.

4

Close to but not exactly equal is

— written with $\approx$.

5

A table whose every column holds the same comparison is a

.

Write About the Math

Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?

¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

☐

Conversion factor

Factor de conversión

☐

Customary system

Sistema usual

☐

Metric system

Sistema métrico

☐

Approximately

Aproximadamente

☐

Ratio table

Tabla de razones

Model: A mile is longer than a kilometer ($1 \text{ km} \approx 0.6 \text{ mi}$), so 5 miles is the longer race even though the numbers match. — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- The conversion factor is $1 \text{ km} \approx \underline{\hspace{1cm}}$ mi. Scaling that up, 100 kilometers $\approx \underline{\hspace{1cm}}$ miles, so the sign means about $\underline{\hspace{1cm}}$ miles per hour. That is $\underline{\hspace{1cm}}$ than 100 miles per hour, because in one hour you would travel $\underline{\hspace{1cm}}$ distance.
- My answer is $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$.

Mi respuesta es $\underline{\hspace{1cm}}$ porque $\underline{\hspace{1cm}}$.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is $\underline{\hspace{1cm}}$.
- My evidence is $\underline{\hspace{1cm}}$, so $\underline{\hspace{1cm}}$.

Mi evidencia es $\underline{\hspace{1cm}}$, entonces $\underline{\hspace{1cm}}$.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.